

IEDA 4000H

Homework 2

Due Date: Oct 12 (Sunday)

1. In Portfolio Optimization, one is interested in optimally allocating a certain amount of money into a set of N assets (stocks), where the portfolio usually denoted via $\mathbf{w} \in \mathbb{R}^N$ is the normalized money invested in each asset (such that $\sum_{i=1}^N w_i = 1$). In Modern Portfolio Theory known after Harry Markowitz, the portfolio design is based on maximizing the return while minimizing the risk. This scheme, also known as Mean-Variance Portfolio (MVP), can be formulated via the following optimization problem:

$$\begin{aligned} \max_{\mathbf{w} \in \mathbb{R}^N} \quad & \boldsymbol{\mu}^\top \mathbf{w} \\ \text{s.t.} \quad & \mathbf{e}^\top \mathbf{w} = 1, \\ & \mathbf{w}^\top \Sigma \mathbf{w} \leq \alpha, \\ & \mathbf{w} \geq 0, \end{aligned} \tag{1}$$

where $\mathbf{e}$ is the vector of ones, $\boldsymbol{\mu}$ is the expected (mean) returns of the stocks, and Σ is the covariance matrix of the stock returns. The term $\mathbf{w}^\top \Sigma \mathbf{w}$ represents the portfolio risk (variance), the square root of which is called volatility. Hence, the objective is to maximize the profit while minimizing the risk, with the parameter α controlling the accepted level of risk. For each solution $\mathbf{w}_\alpha^*$, we define the following quantities:

- Portfolio Expected Return: $\text{PER}(\alpha) = \boldsymbol{\mu}^\top \mathbf{w}_\alpha^*$,
- Portfolio Volatility: $\text{PV}(\alpha) = \sqrt{\mathbf{w}_\alpha^{*\top} \Sigma \mathbf{w}_\alpha^*}$,
- Sharpe Ratio: $\text{SR}(\alpha) = \frac{\text{PER}(\alpha)}{\text{PV}(\alpha)}$.

(a) Argue that whether (1) is a convex optimization problem or not.

(b) Let

$$\boldsymbol{\mu} = \begin{pmatrix} 0.06 \\ 0.01 \\ 0.03 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 1.0 & 0.02 & -0.01 \\ 0.02 & 1.0 & -0.1 \\ -0.01 & -0.1 & 1.0 \end{pmatrix}.$$

Use CVXPY to solve (1) for $\alpha \in \{4, 6, \dots, 14\} \times 10^{-1}$. Plot the efficient frontier, i.e. the diagram of $\text{PER}(\alpha)$ versus $\text{PV}(\alpha)$.

(c) For which value of α in (b), is the Sharpe Ratio maximized?

2. Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a closed convex set, θ and f be convex functions on $\mathbb{R}^n$. Suppose that θ is not necessarily smooth. Let f be differentiable on some open set that contains $\mathcal{X}$. Assume that the solution set of the minimization problem $\min\{\theta(\mathbf{x}) + f(\mathbf{x}) \mid \mathbf{x} \in \mathcal{X}\}$ is nonempty. Show that

$$\mathbf{x}^* \in \arg \min\{\theta(\mathbf{x}) + f(\mathbf{x}) \mid \mathbf{x} \in \mathcal{X}\} \quad (2a)$$

if and only if

$$\mathbf{x}^* \in \mathcal{X}, \quad \theta(\mathbf{x}) - \theta(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (2b)$$

Hint: Suppose that (2a) holds. Define $\mathbf{x}_\alpha = (1 - \alpha)\mathbf{x}^* + \alpha\mathbf{x}$ for any $\alpha \in (0, 1]$. Then, for any $\mathbf{x} \in \mathcal{X}$, we have the following inequality:

$$\frac{\theta(\mathbf{x}_\alpha) - \theta(\mathbf{x}^*)}{\alpha} + \frac{f(\mathbf{x}_\alpha) - f(\mathbf{x}^*)}{\alpha} \geq 0.$$

3. Consider the following optimization problem:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \mathbf{x}^\top \mathbf{W} \mathbf{x} \\ \text{s.t.} \quad & x_i^2 = 1, \quad i = 1, \dots, n, \end{aligned} \quad (3)$$

where $\mathbf{W} \in \mathbb{R}^{n \times n}$ is a symmetric matrix. Let $\mathbf{e}$ denote the vector of all ones and for $\boldsymbol{\lambda} \in \mathbb{R}^n$,

$$\text{diag}(\boldsymbol{\lambda}) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

- (a) Write the Lagrangian function of (3).
- (b) What is the Lagrangian dual of (3)?
- (c) Let $\mathbf{W} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Solve (3) and its Lagrangian dual problem. Find the duality gap.